

Chapter 18

Integrals

Only One Option Correct Type Questions

1. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals

[IIT-JEE-2007 (Paper-2)]

- (A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}} + K$
(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}} + K$

2. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant C , the value of $J - I$ equals

[IIT-JEE-2008 (Paper-2)]

- (A) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + C$ (B) $\frac{1}{2} \log \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + C$
(C) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$ (D) $\frac{1}{2} \log \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + C$

3. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)

[IIT-JEE-2012 (Paper-1)]

- (A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
(B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
(D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

Paragraph for Q.No. 4

Choose the correct answer :

Let $\psi_1 : [0, \infty) \rightarrow \mathbb{R}$, $\psi_2 : [0, \infty) \rightarrow \mathbb{R}$, $f : [0, \infty) \rightarrow \mathbb{R}$ and $g : [0, \infty) \rightarrow \mathbb{R}$ be functions such that $f(0) = g(0) = 0$,

$$\psi_1(x) = e^{-x} + x, x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt, x > 0$$

$$\text{and } g(x) = \int_0^{x^2} \sqrt{t} e^{-t} dt, x > 0.$$

[JEE (Adv)-2021 (Paper-2)]

4. Which of the following statements is TRUE?

(A) $\Psi_1(x) \leq 1$, for all $x > 0$

(B) $\Psi_2(x) \leq 0$, for all $x > 0$

(C) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, for all $x \in \left(0, \frac{1}{2}\right)$

5. For positive integer n , define $f(n) = n + \frac{16+5n-3n^2}{4n+3n^2} + \frac{32+n-3n^2}{8n+3n^2} + \frac{48-3n-3n^2}{12n+3n^2} + \dots + \frac{25n-7n^2}{7n^2}$.

Then, the value of $\lim_{n \rightarrow \infty} f(n)$ is equal to

(A) $3 + \frac{4}{3} \log_e 7$

(B) $4 - \frac{3}{4} \log_e \left(\frac{7}{3}\right)$

(C) $4 - \frac{4}{3} \log_e \left(\frac{7}{3}\right)$

(D) $3 + \frac{3}{4} \log_e 7$

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